

8300100875
VBH0310CSPFS

EC centrifugal module - RadiPac

backward-curved, single-intake
with support bracket

ebm-papst Mulfingen GmbH & Co. KG
Bachmühle 2 · D-74673 Mulfingen
Phone +49 7938 81-0
Fax +49 7938 81-110
info1@de.ebmpapst.com
www.ebmpapst.com

Limited partnership · Headquarters Mulfingen
Amtsgericht (court of registration) Stuttgart · HRA 590344

General partner Elektrobau Mulfingen GmbH · Headquarters Mulfingen
Amtsgericht (court of registration) Stuttgart · HRB 590142

Nominal data

Item	8300100875	
Motor	E09002-28	
Phase		1~
Nominal voltage	VAC	230
Nominal voltage range	VAC	200 .. 277
Frequency	Hz	50/60
Method of obtaining data		ml
Speed (rpm)	min ⁻¹	2590
Power consumption	W	500
Current draw	A	2.2
Min. ambient temperature	°C	-40
Max. ambient temperature	°C	60

ml = Max. load · me = Max. efficiency · fa = Free air · cs = Customer specification · ce = Customer equipment
Subject to change

Data according to Commission Regulation (EU) 327/2011 (prEN 17166)

		Actual	Req. 2015			
01 Overall efficiency η_{es}	%	68.8	48.3	09 Power consumption P_{ed}	kW	0.49
02 Measurement category	A			09 Air flow q_v	m³/h	2495
03 Efficiency category	Static			09 Pressure increase p_{fs}	Pa	448
04 Efficiency grade N		82.5	62	10 Speed (rpm) n	min ⁻¹	2590
05 Variable speed drive		Yes		11 Specific ratio*		1.00

Data obtained at optimum efficiency level.

The efficiency values displayed for achieving conformity with the Ecodesign Regulation EU 327/2011 has been reached with defined air duct components (e.g. inlet rings). The dimensions must be requested from ebm-papst. If other air conduction geometries are used on the installation side, the ebm-papst evaluation loses its validity/the conformity must be confirmed again. The product does not fall within the scope of Regulation (EU) 2019/1781 due to the exception specified in Article 2 (2a) (motors completely integrated into a product).

* Specific ratio = $1 + p_b / 100\,000\text{ Pa}$ LU-224424

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Technical description

Size	310 mm
Motor size	90
Rotor surface	Painted black
Electronics housing material	Die-cast aluminum
Impeller material	PP plastic
Support plate material	Sheet steel, galvanized
Support bracket material	Steel, painted black
Inlet nozzle material	ABS plastic
Number of blades	5
Direction of rotation	Clockwise, viewed toward rotor
Degree of protection	IP55
Insulation class	"F"
Moisture (F) / Environmental (H) protection class	H1
Ambient temperature note	Occasional start-up at temperatures between -40°C and -25°C is permitted. For continuous operation at ambient temperatures below -25°C (such as refrigeration applications), use must be made of a fan design with special low-temperature bearings.
Max. permitted ambient temp. for motor (transport/storage)	+80 °C
Min. permitted ambient temp. for motor (transport/storage)	-40 °C
Installation position	See legend on product drawing
Condensation drainage holes	On rotor side
Mode	S1
Motor bearing	Ball bearing
Technical features	- Output 10 VDC, max. 10 mA - Operation and alarm display with LED - Locked-rotor detection - Speed control - External 15-30 VDC input (parameterization) - Alarm relay - Configurable inputs/outputs (I/O) - Power limiter - MODBUS V7.0 - Motor current limitation - PFC, active - RS-485 MODBUS-RTU - Soft start - Control interface with SELV potential safely disconnected from the mains - Temperature derating - Thermal overload protection for electronics/motor - Line undervoltage / phase failure detection
Touch current according to IEC 60990 (measuring circuit Fig. 4, TN system)	<= 3.5 mA
Motor protection	Electronic motor protection
With cable	Lateral

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Protection class assignment	I; If a protective earth is connected by the customer This component for installation may have several local protection classes. This information relates to this component's basic design. The final protection class is based on the component's intended installation and connection.
Conformity with standards	EN 60335-1; EN 61800-5-1; UKCA; CE
Approval	UL 1004-7 + 60730-1; CSA C22.2 No. 77 + CAN/CSA-E60730-1

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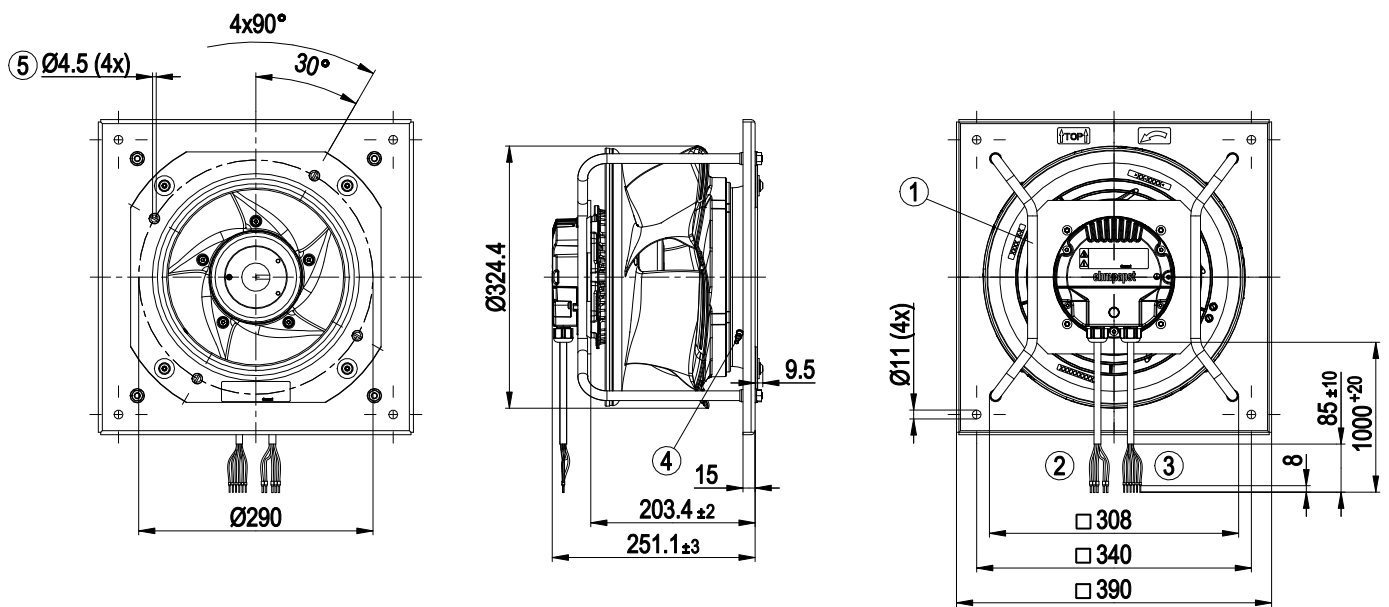
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Product drawing



1	Installed position: shaft horizontal (install support struts only vertically as illustrated) or rotor on bottom; rotor on top on request
2	Supply line (PWR) PVC AWG18 5x wire-end ferrule
3	Control wire (CTRL) PVC AWG22 6x wire-end ferrule
4	Inlet ring with pressure tap (k-factor: 115)
5	Fastening holes for FlowGrid 25310-2-2957 (not included in scope of delivery) are provided and must be subsequently opened as required

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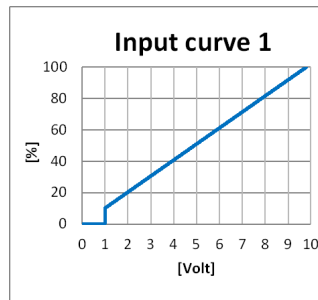
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Connection diagram



No.	Conn.	Designation	Color	Function/assignment
	PWR	PE	green/yellow	Protective earth
	PWR	L	black	Power supply, phase, see nameplate for voltage range
	PWR	N	blue	Power supply, neutral conductor, see nameplate for voltage range
	PWR	COM	orange	Status relay, floating status contact, common connection, contact rating 250 VAC / 2 A (AC1) / min. 10 mA, reinforced insulation on supply side and on control interface side
	PWR	NC	orange	Status relay, floating status contact, break for failure
	CTRL	GND	blue	Reference ground for control interface, SELV
	CTRL	Vout	red	Voltage output 10 VDC +/-3%, I _{max} =10 mA Short-circuit-proof, power supply for external devices, SELV alternative: 15-30 VDC input for parameterization via MODBUS without line voltage
	CTRL	IO1	yellow	Factory setting: Analog input 0-10 V / PWM, R _i =100 kΩ, function: set value Characteristic curve parameterizable (see input characteristic curve "Input curve 1"), SELV Function parameterizable (see table Optional interface functions)
	CTRL	IO2	white	Factory setting: Open collector output, U _{max} =50 VDC, I _{max} =20 mA, function:Tacho output 1 pulse/revolution, SELV Function parameterizable (see table Optional interface functions)
	CTRL	RSA	gray	RS-485 interface for MODBUS RSA, SELV dielectric strength to MODBUS RSB +/-14 V, dielectric strength to GND +/-7 V
	CTRL	RSB	brown	RS-485 interface for MODBUS RSB, SELV dielectric strength to MODBUS RSA +/-14 V, dielectric strength to GND +/-7 V
		LED		green: status = good, ready for operation orange: status = warning red: status = failure

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Terminal/plug assignment

Medium (M2)		Functions and parameter description MODBUS V7.0		configurable IO mode		electrical specification		MODBUS Register for IO mode configuration		configurable IO functions: normal / inverse	
CON2	configurable IO mode										
IO1	Dim1 (active high): digital input	active: applied voltage 3,5-50 VDC, SELV not active: pin open or applied voltage < 1,5 VDC SELV	D158 [0]								
	Dim1 (active low): digital input	active: applied voltage < 1,5 VDC, SELV not active: pin open or applied voltage 3,5-50 VDC SELV	D158 [1]								
	Ain1 0-10 V/PWM: analog input	RI = 100 kΩ, characteristic curve parameterizable, $f_{PWM} = 1k..10 kHz$, SELV	D158 [2]								
	Ain1 0-10 V/PWM (with pull up): analog input	RI = 100 kΩ, characteristic curve parameterizable, $f_{PWM} = 1k..10 kHz$, SELV	D158 [3]								
	Dim1 (active high): digital input	active: applied voltage 3,5-50 VDC, SELV not active: pin open or applied voltage < 1,5 VDC SELV	D159 [0]								
IO2	Ain1 0-10 V/PWM: analog input	RI = 100 kΩ, characteristic curve parameterizable, $f_{PWM} = 1k..10 kHz$, SELV	D159 [2]								
	Tach out (open collector)	Umax=50 VDC, Imax=20 mA, SELV	D159 [5]								
	Diagnostics out (open collector)	Umax=50 VDC, Imax=20 mA, SELV	D159 [6]								
	Alarm out (open collector)	Umax=50 VDC, Imax=20 mA, SELV	D159 [10]								
	Test pulse output addressing (open collector)	Umax=50 VDC, Imax=20 mA, SELV	D159 [11]								
RSA RSB	DCI-output (open collector)	Umax=50 VDC, Imax=20 mA, SELV	D159 [12]								
COM NC	RS485 bus connection	MODBUS RTU, parameter specification V7, SELV									
Vout	Relay	250 VAC / 2 A (AC1)									
	Voltage output	Voltage 10 VDC, SELV									
	Alternatively: Input auxiliary power supply for parameterization via RS485/MODBUS RTU without line voltage	15...30 VDC									
selection direct via IO											
signal: tach out											
signal: diagnostics out											
signal: alarm out											
signal: output pulses autoaddressing											
signal: DCI out											
signal: fan modulation level %											
signal: actual speed 1/min											
signal: system modulation level %											
signal: remote control output 0-10 V											
signal: run monitoring											
signal: status											

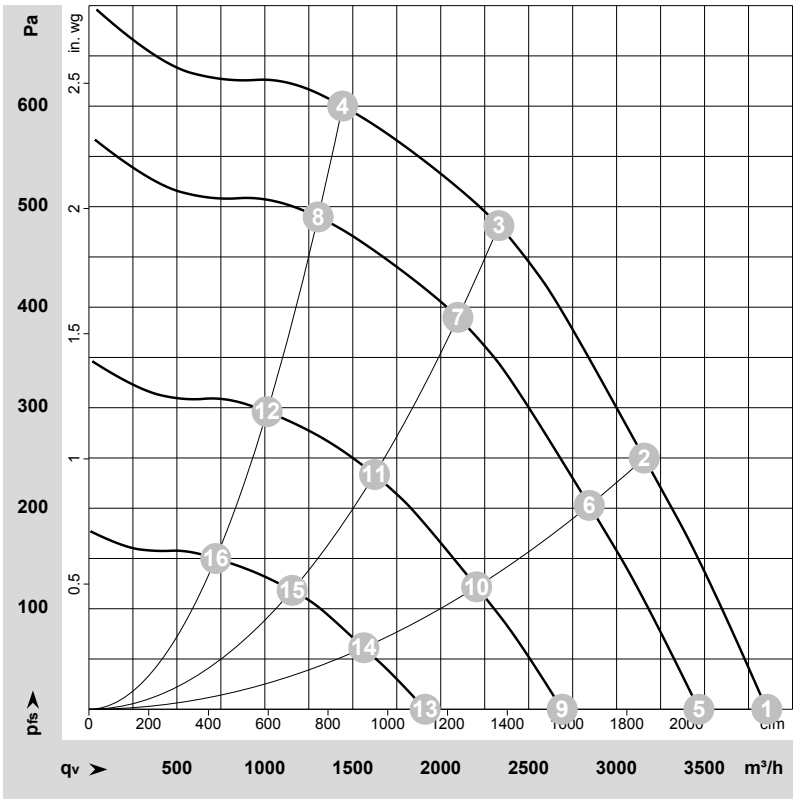
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Curves: Air performance 50 Hz



$\rho = 1.15 \text{ kg/m}^3 \pm 2 \%$

Measurement: LU-224424-1
Date: 2022-12-08
Nozzle: 8217101930

Air performance measured according to ISO 5801 installation category A. For detailed information on the measurement setup, contact ebm-papst. Intake sound level: Sound power level according to ISO 13347 / sound pressure level measured at 1 m distance from fan axis. The values given are valid under the specified measuring conditions and may vary due to conditions of installation. For deviations from the standard configuration, the parameters have to be checked on the installed unit.

Measured values

	Wired	U	f	n	P _e	I	LpA _{in}	LwA _{in}	LwA _{out}	LwA	q _v	P _{fs}	q _v	P _{fs}
		V	Hz	min ⁻¹	W	A	dB(A)	dB(A)	dB(A)	dB	m³/h	Pa	cfm	in. wg
1	1~	230	50	2590	353	1.57	73	81	85	87	3855	0	2270	0.00
2	1~	230	50	2590	432	1.91	66	74	79	80	3160	250	1860	1.00
3	1~	230	50	2590	500	2.20	63	70	77	77	2330	480	1370	1.93
4	1~	230	50	2590	471	2.08	68	75	79	81	1440	600	850	2.41
5	1~	230	50	2335	262	1.18	70	79	82	84	3470	0	2040	0.00
6	1~	230	50	2335	320	1.43	63	71	76	78	2845	203	1675	0.81
7	1~	230	50	2335	370	1.65	61	68	74	75	2100	390	1235	1.57
8	1~	230	50	2335	349	1.55	65	72	76	78	1305	491	765	1.97
9	1~	230	50	1820	134	0.63	64	72	76	77	2690	0	1585	0.00
10	1~	230	50	1815	160	0.74	57	66	70	71	2205	122	1295	0.49
11	1~	230	50	1815	183	0.83	55	62	68	69	1625	234	955	0.94
12	1~	230	50	1815	174	0.79	58	65	69	71	1015	297	595	1.19
13	1~	230	50	1295	59	0.34	57	65	69	70	1910	0	1125	0.00
14	1~	230	50	1300	68	0.37	50	58	63	64	1565	61	920	0.24
15	1~	230	50	1295	76	0.40	48	55	60	61	1155	118	680	0.47
16	1~	230	50	1300	73	0.39	49	56	60	62	720	150	425	0.60

Wired = Wiring · U = Voltage · f = Frequency · n = Speed (rpm) · P_e = Power consumption · I = Current draw · LpA_{in} = Sound pressure level intake side · LwA_{in} = Sound power level intake side
LwA_{out} = Sound power level outlet side · q_v = Air flow · P_{fs} = Pressure increase